Chapter 15 Review Paper

Hardware, Boolean algebra and logic circuits

练习卷，不是 Cambridge 原卷。90 分钟，75 分。做的时候不要打开 ms.html。范围说明在 syllabus.html。

Time	1 hour 30 minutes
Total marks	75
Syllabus	15.1 Processors and parallel processing 15.2 Boolean algebra and logic circuits
Materials	None. Write on this paper.

Instructions

1. Answer **all** questions.
2. Show the law used at each stage when a question asks for working.
3. A bar, an apostrophe, or the word NOT all mean NOT. A dot means AND. + means OR. ⊕ means XOR.
4. The number of marks is given in brackets [ ].

Question	Marks
1 RISC, CISC, pipelining and interrupts	12
2 Parallel architectures	12
3 Logic circuits and adders	15
4 Flip-flops	14
5 De Morgan’s laws and Boolean algebra	10
6 Karnaugh maps	12
Total	**75**

1 Processors [12]

(a) Tick (☐) one box in each row to show whether the statement describes a RISC processor or a CISC processor.

One mark is given for every two correct rows. [3]

Statement	RISC	CISC
An instruction is normally executed in one clock cycle.	☐	☐
The control unit is microprogrammed.	☐	☐
The processor has a large number of general-purpose registers.	☐	☐
Instructions can have different lengths.	☐	☐
Memory is accessed only by load and store instructions.	☐	☐
The instruction set provides many addressing modes.	☐	☐

(b) Describe how pipelining is used during the fetch-execute cycle in a RISC processor. [3]

(c) Explain how interrupt handling on a pipelined RISC processor differs from interrupt handling on a CISC processor that does not use pipelining. [4]

(d) Explain why a RISC processor uses a large number of general-purpose registers. [2]

2 Parallel processing [12]

(a) Complete the table with the full name of each computer architecture. [2]

One mark is given for every two correct rows.

Abbreviation	Full name
SISD	
SIMD	
MISD	
MIMD	

(b) Three computer systems are described below. Identify the architecture used by each system. Use one of SISD, SIMD, MISD or MIMD. [3]

System 1 has one processor. The processor carries out one instruction, on one data value, and then starts the next instruction.

Architecture .....

System 2 has 64 arithmetic units. In one clock cycle every unit carries out the same add instruction. Each unit receives a different pair of numbers.

Architecture .....

System 3 has four processors and one shared memory. At one moment, processor 1 is executing a divide, processor 2 is executing a branch, and processors 3 and 4 are each executing a different instruction on their own data.

Architecture .....

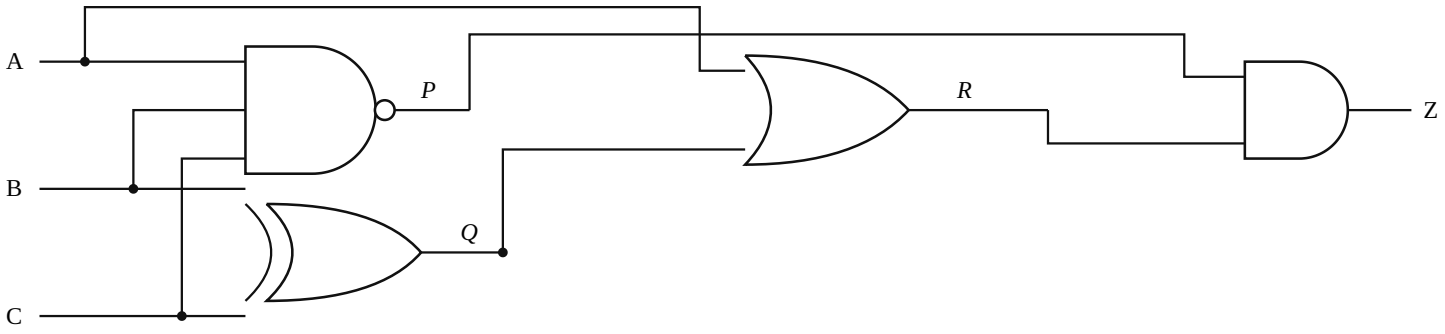
(c) Describe the MISD architecture. State why it is rarely used in a general-purpose computer. [3]

(d) State what is meant by a massively parallel computer. [1]

(e) A laboratory uses a massively parallel computer to process one very large set of experimental results. Explain how the processors work together on that single task. [3]

3 Logic circuits and adders [15]

The diagram shows a logic circuit. P, Q and R are intermediate outputs. Z is the output of the circuit. A dot on a wire is a junction. Crossing wires do not join.



(a) (i) Complete the truth table for this logic circuit. Show your working in columns P, Q and R. [4]

Inputs			Working space			Output
A	**B**	**C**	**P**	**Q**	**R**	**Z**
0	0	0				
0	0	1				
0	1	0				
0	1	1				
1	0	0				
1	0	1				
1	1	0				
1	1	1				

(a) (ii) Write the Boolean expression for Z as a sum-of-products. Do not simplify the expression. [2]

Z = ...

A different circuit adds two one-bit binary numbers and has two outputs.

(b) State the name of this circuit, and state the purpose of each output. [3]

Name .....

First output .....

Second output .....

(c) Explain why that circuit cannot, on its own, add two multi-bit binary numbers. Name the circuit that removes the problem. [2]

(d) A full adder has inputs A, B and C_{in}. Write the Boolean expression for the sum output and for the carry-out output. [2]

Sum = ...

C_{out} = ...

(e) A full adder can be made from two half adders and one other gate. Name that gate, and state what its two inputs are. [2]

Gate .....

Inputs .....

4 Flip-flops [14]

(a) Define a combinational circuit and a sequential circuit. [2]

Combinational .....

Sequential .....

(b) Draw a logic circuit for an SR flip-flop made from two NOR gates. Label the inputs S and R, and the outputs Q and Q'. Q' means NOT Q. Show the cross-coupling. [3]

(c) Complete the table for the NOR SR flip-flop you drew in part (b). [4]

S	R	Q	Q'
0	0		
0	1		
1	0		
1	1		

(d) Explain why the input combination S = 1, R = 1 is not used. [2]

(e) State the purpose of a flip-flop in a processor. [1]

(f) A JK flip-flop is clocked. State the effect on Q when J = 1 and K = 1, and state why this input combination does not cause the problem identified in part (d). [2]

5 Boolean algebra [10]

(a) State De Morgan's two laws. [2]

Law 1 .....

Law 2 .....

(b) Simplify the following expression using De Morgan's laws and Boolean algebra. Show every stage and name the law used at each stage. [6]

X = NOT(A . NOT(B + C)) . (B + A . NOT B)

X = ...

(c) Rewrite your answer to part (b) so that the expression can be built from NAND gates only. Both inputs of a NAND gate may be connected to the same signal. Show your working. [2]

X = ...

6 Karnaugh maps [12]

(a) State what is meant by a sum-of-products expression. [1]

(b) Describe one benefit of using a Karnaugh map to simplify a Boolean expression. [1]

(c) The column headings of a Karnaugh map are written 00, 01, 11, 10 and not 00, 01, 10, 11. Explain why. [2]

A Boolean function X has the truth table below.

A	B	C	D	X
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1

(d) Complete the Karnaugh map for X. [2]

CD \ AB	00	01	11	10
00				
01				
11				
10				

(e) Draw loops around the groups of 1s that produce an optimal sum-of-products. Draw the loops on the map in part (d). [3]

(f) Write the simplified sum-of-products for X from your loops. Do not factorise it yet. [2]

X = ...

(g) Use Boolean algebra to write your answer to part (f) in its simplest form. [1]

X = ...

End of paper. Total: 75 marks.